Project 2 Database Design Document

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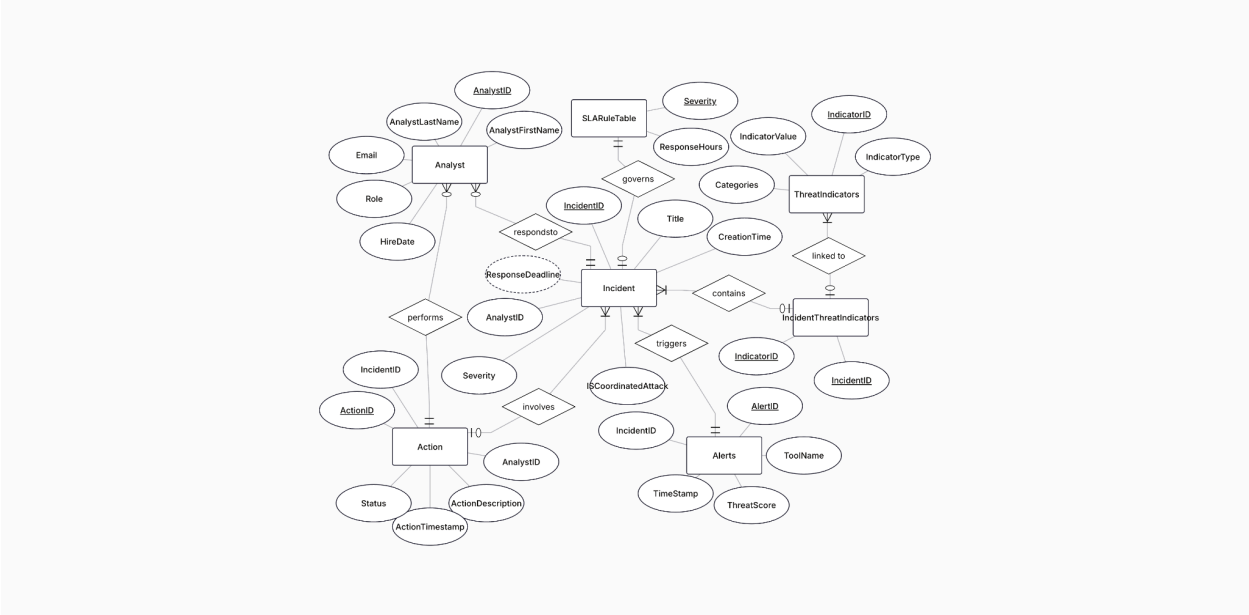
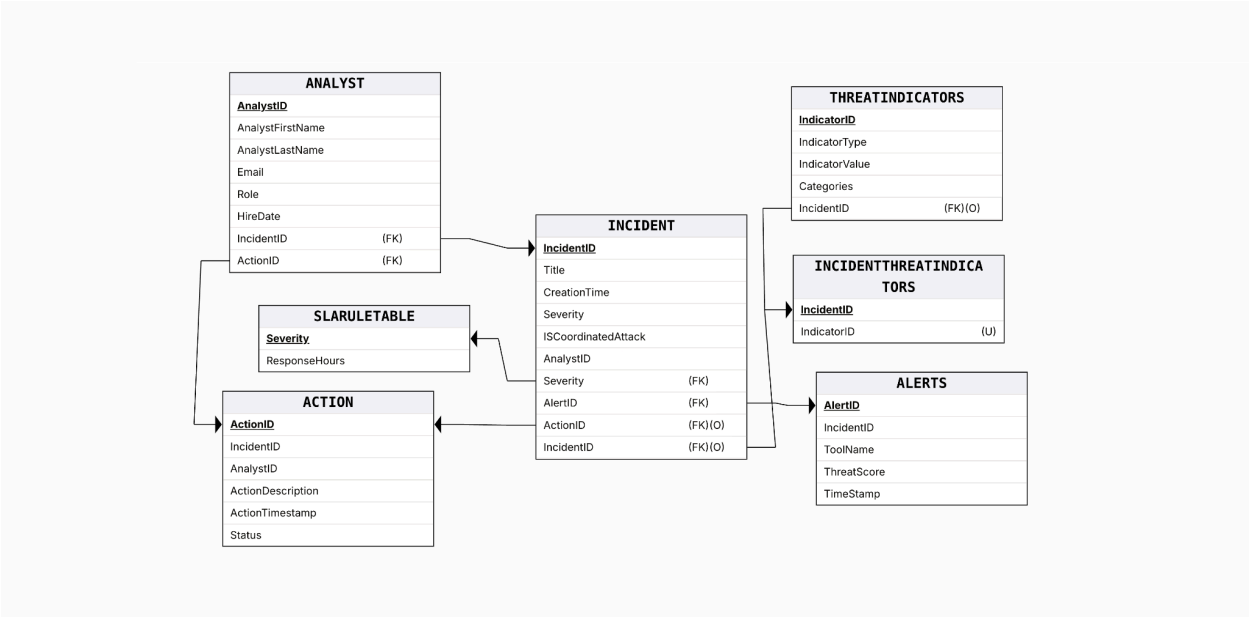
ER Diagram



Relational Schema



Assumptions

Project 2 Assumptions

1. Each incident is assigned to exactly one primary analyst who is responsible for the initial response, even though multiple analysts may perform actions related to that incident.
2. An analyst can work on multiple incidents and can perform multiple actions across different incidents.
3. Each alert is associated with exactly one incident, but an incident may have multiple alerts generated from different monitoring tools.
4. Each action is linked to exactly one incident and is performed by exactly one analyst.
5. An incident may have multiple actions, but actions are recorded independently and are not required to follow a strict sequence.
6. Threat indicators such as IP addresses, domains, and file hashes may appear in multiple incidents, and each incident may contain multiple threat indicators, creating a many to many relationship.
7. The IndicatorType field is restricted to only three valid values: ‘IP Address’, ‘Domain’, and ‘File Hash’.
8. The Categories field in ThreatIndicators contains a single classification (e.g., Ransomware, Phishing, Malware), even though in real systems an indicator could belong to multiple categories.
9. The ResponseDeadline for each incident is derived based on the CreationTime and the SLA rule associated with its severity level.
10. SLA rules are fixed and predefined as:
    - Critical: 1 hour
    - High: 4 hours
    - Medium: 24 hours
    - Low: 48 hours
11. If no action is recorded before the ResponseDeadline, the incident is considered a delayed response for reporting purposes.
12. The Status field in the Action table represents the current state of the action and is limited to ‘Pending’, ‘In Progress’, or ‘Complete’.
13. The system assumes all timestamps are stored in the same time zone and no time zone conversions are required.
14. Each table uses a surrogate primary key (e.g., AnalystID, IncidentID) to uniquely identify records.
15. Data loaded into the system is assumed to be clean, valid, and does not violate referential integrity constraints.